Lab Programs

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30. Bar Plot of Scores by Group and Gender

Aim

To write a Python program to create a bar plot showing scores for men and women using multiple X values on the same chart.

Code

```
import matplotlib.pyplot as plt

import numpy as np


means_men = (22, 30, 35, 35, 26)

means_women = (25, 32, 30, 35, 29)


groups = np.arange(5)

width = 0.35


plt.bar(groups - width/2, means_men,

        width, label="Men", color="blue")


plt.bar(groups + width/2, means_women,

        width, label="Women", color="orange")


plt.xlabel("Groups")

plt.ylabel("Scores")

plt.title("Scores by Group and Gender")
```

plt.xticks(groups, ["Group 1", "Group 2",

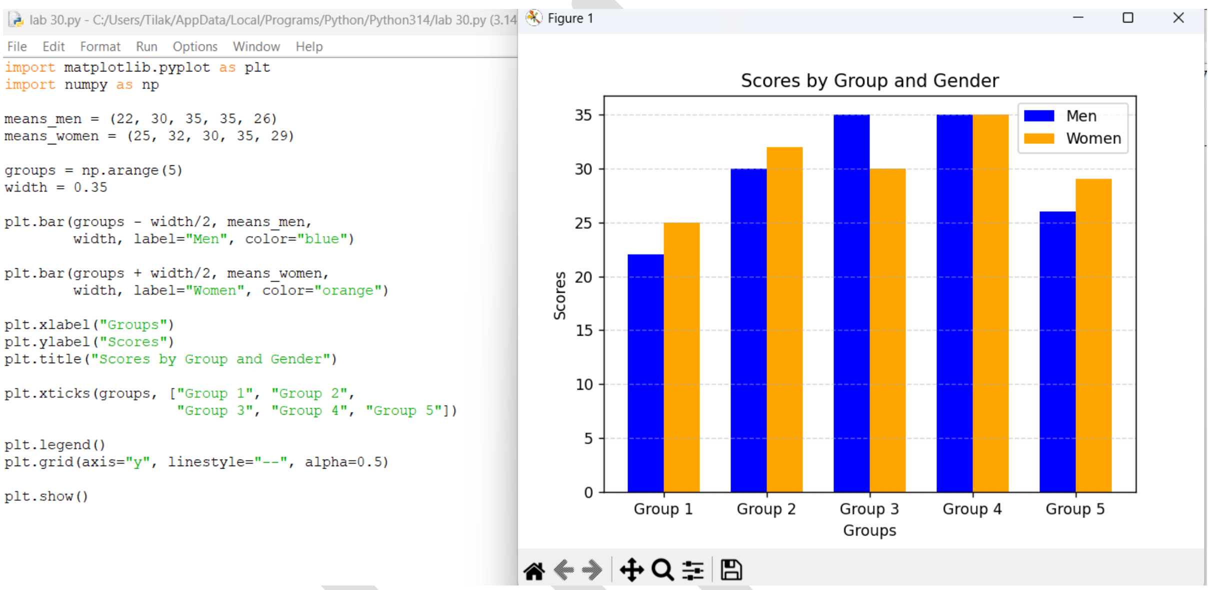
"Group 3", "Group 4", "Group 5"])

plt.legend()

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()

Output



31. Stacked Bar Plot with Error Bars

Aim

To write a Python program to create a stacked bar plot with error bars for men's and women's scores.

Code

import matplotlib.pyplot as plt

import numpy as np

means_men = (22, 30, 35, 35, 26)

means_women = (25, 32, 30, 35, 29)

std_men = (4, 3, 4, 1, 5)

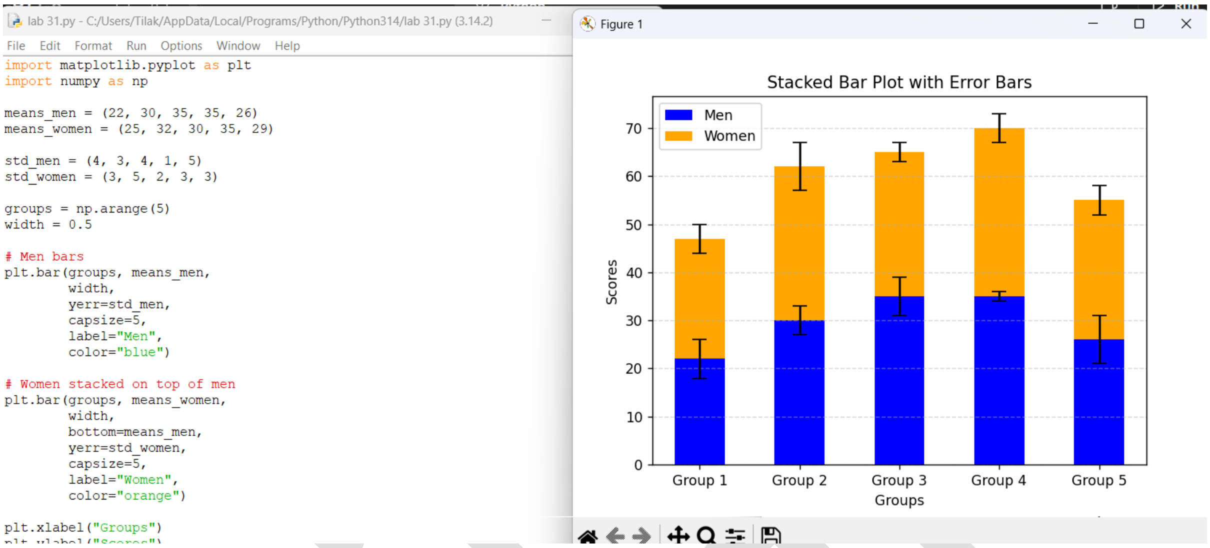
```
std_women = (3, 5, 2, 3, 3)

groups = np.arange(5)
width = 0.5

# Men bars
plt.bar(groups, means_men,
        width,
        yerr=std_men,
        capsize=5,
        label="Men",
        color="blue")

# Women stacked on top of men
plt.bar(groups, means_women,
        width,
        bottom=means_men,
        yerr=std_women,
        capsize=5,
        label="Women",
        color="orange")

plt.xlabel("Groups")
plt.ylabel("Scores")
plt.title("Stacked Bar Plot with Error Bars")

plt.xticks(groups, ["Group 1", "Group 2",
                    "Group 3", "Group 4", "Group 5"])
```

plt.legend()

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()

Output:



32. Scatter Graph with Random Distribution

Aim

To write a Python program to generate random X and Y values and draw a scatter graph by plotting them against each other.

Code

```
import matplotlib.pyplot as plt

import numpy as np


# Generate random values

np.random.seed(10)


x = np.random.rand(50)

y = np.random.rand(50)


# Draw scatter plot
```

```
plt.scatter(x, y,
            color="blue",
            marker="o",
            alpha=0.7)

plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Scatter Plot of Random Distribution")

plt.grid(True, linestyle="--", alpha=0.5)

plt.show()
```

Output: